# Hotel Revenue Analysis Dashboard

## Infosys Springboard Internship Project

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# Identifying the Business Challenge

Hotels accumulate vast amounts of booking data, yet often struggle with a lack of clear visibility into their primary revenue drivers. This challenge extends to analyzing seasonal performance trends and establishing a structured system for overall performance tracking. There is a critical need for a data-driven decision-making dashboard.

### Data Overload, Limited Insight

Hotels generate large volumes of booking data without clear insights into key performance indicators.

### Seasonal Performance Gaps

Difficulty in analyzing and understanding seasonal revenue and occupancy trends.

### Absence of Centralized Tracking

No structured system to consistently monitor and evaluate overall hotel performance.

# Project Objectives: Enhancing Data-Driven Decisions

This project aimed to transform raw hotel booking data into actionable intelligence through a comprehensive Power BI dashboard. The core objectives included data preparation, model construction, KPI definition, and interactive dashboard design.

01

## Clean & Prepare Data

Raw hotel booking data was meticulously cleaned and prepared for analysis.

02

## Build Structured Data Model

A robust star schema data model was developed to ensure data integrity and query efficiency.

03

## Define Key Performance Indicators

Critical KPIs were created using DAX formulas to measure and monitor hotel performance.

04

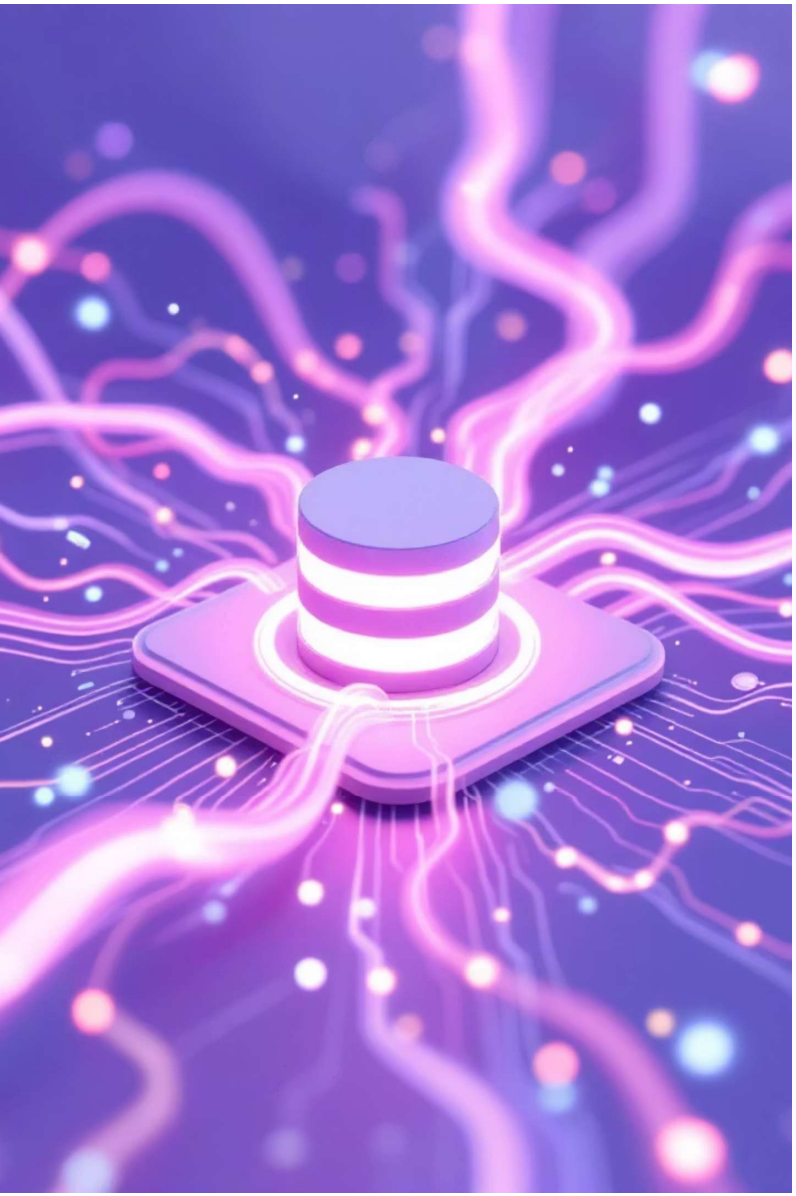
## Analyze Revenue & Bookings

Deep-dive analysis into revenue and booking performance patterns.

05

## Design Interactive Dashboard

A 3-page interactive Power BI dashboard was designed for intuitive data exploration.



# Dataset Overview: The Foundation of Our Analysis

The analysis was built upon a rich dataset of hotel booking records, encompassing various critical details. Understanding these columns was fundamental to extracting meaningful insights and building a comprehensive dashboard.

## Key Data Points:

- Arrival Dates
- ADR (Average Daily Rate)
- Stays (Weekend/Week Nights)
- Market Segment
- Country of Origin
- Hotel Type (City/Resort)

## Purpose:

This dataset provided the raw material for understanding booking patterns, revenue generation, and market behavior. Each column offered unique insights crucial for the analytical objectives.

The dataset underwent rigorous cleaning and transformation to ensure accuracy and reliability for subsequent analysis.

# Data Cleaning & Preparation: Refining Raw Data

Before any meaningful analysis could begin, the raw hotel booking data required extensive cleaning and preparation. This phase was crucial for removing inconsistencies, correcting errors, and creating calculated columns essential for KPI derivation.

1

### Column Management

Removed null values and irrelevant columns to streamline the dataset.

2

### ADR Validation

Filtered out invalid Average Daily Rate (ADR) values where ADR was less than or equal to zero.

3

### Data Type Correction

Ensured appropriate data types (Date, Decimal, Whole Number) for all relevant columns.

4

### Feature Engineering

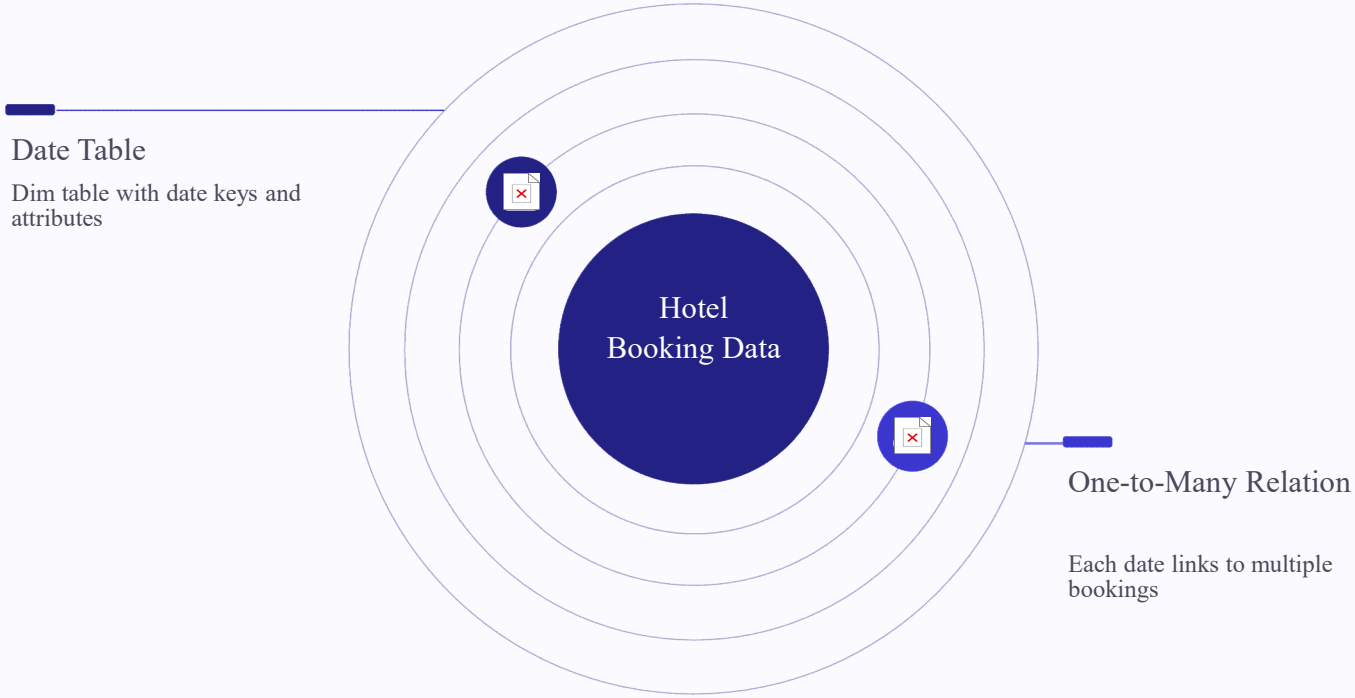
Created "Total Nights" (Weekend + Week Nights) and "Total Revenue" (ADR × Total Nights).

5

### Date Table Integration

Developed a separate, dedicated Date Table to facilitate robust time-based analysis.

# Data Modeling: Building a Star Schema




A star schema was implemented to create a structured and efficient data model. This approach ensures data integrity, optimizes query performance, and forms a professional foundation for business intelligence reporting.

### Fact Table: Hotel Booking Data

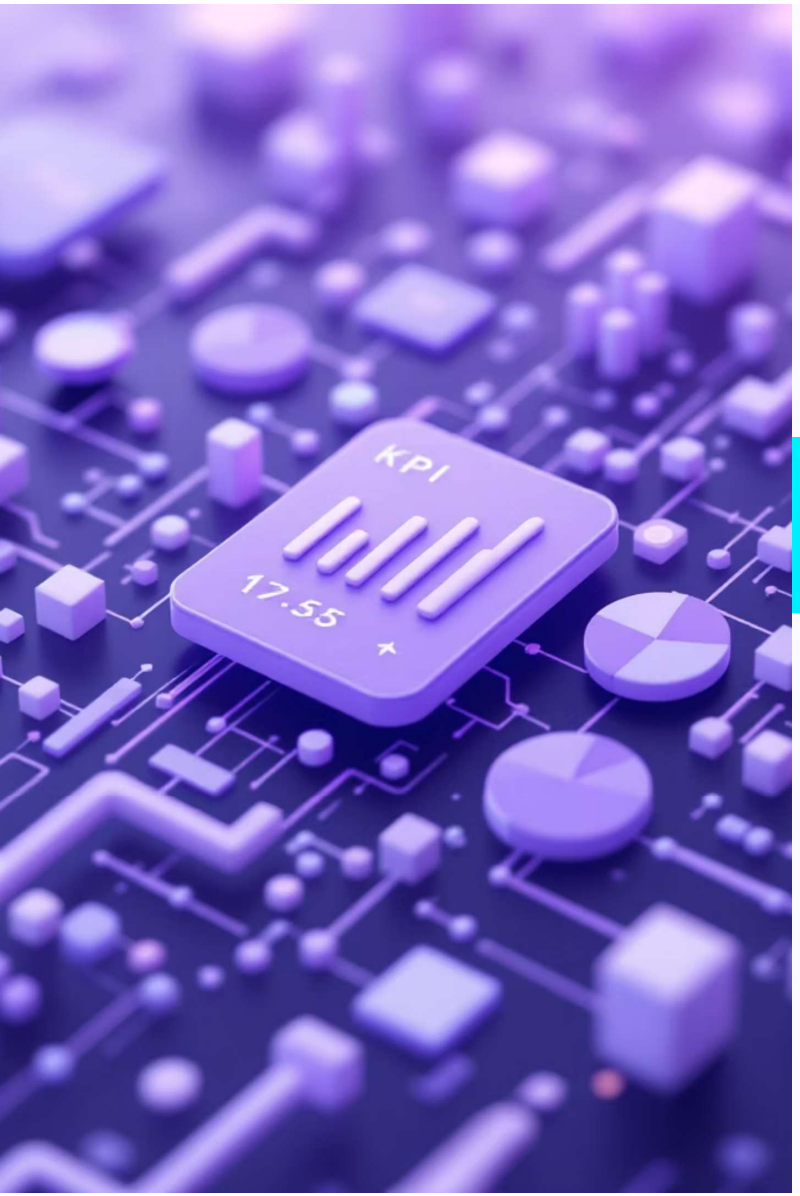
Central table containing core booking metrics.

### Dimension Table: Date Table

Provides chronological context for time-series analysis.

### Relationship: One-to-Many

Date (dimension) to Hotel Booking Data (fact), enabling accurate time-based filtering.




# Key Performance Indicators (KPIs): Measuring Success with DAX

Key Performance Indicators (KPIs) are the pulse of business performance. Utilizing Data Analysis Expressions (DAX) in Power BI, several crucial KPIs were developed to provide a comprehensive view of hotel operations and financial health.

| Net Revenue | Average ADR | Total Bookings | Avg. Days of stay | Guests in Hotels | Avg. Days of stay |
|---|---|---|---|---|---|
| 30.5M | 100.8 | 86.8K | 3.5 | 168.7K | 351.7 |

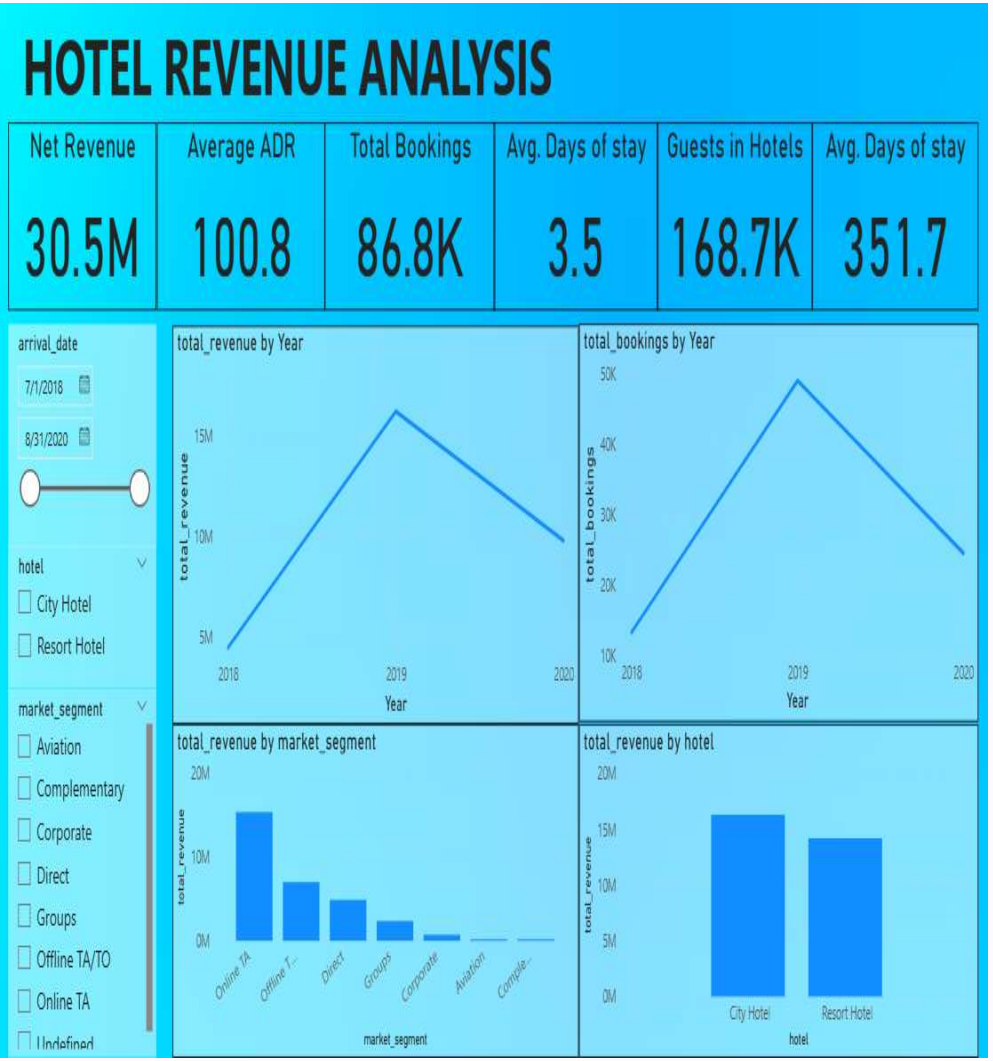
## Core Revenue Metrics:

- **Total Revenue:** SUM(Total Revenue)
- **Average ADR:** AVERAGE(adr)

## Booking & Occupancy:

- **Total Bookings:** COUNTROWS(Hotel)
- **Total Nights:** SUM(Total Nights)

These KPIs form the backbone of the dashboard, allowing management to track, analyze, and make informed decisions on hotel performance.




# Dashboard Page 1: Executive Summary

The Executive Summary page offers a high-level overview of the hotel's performance, providing top management with critical insights at a glance. This page is designed for quick performance assessment and strategic decision-making.

**30.5M**

Net Revenue

Overall financial performance

**100.8**

Average ADR

Average Daily Rate achieved

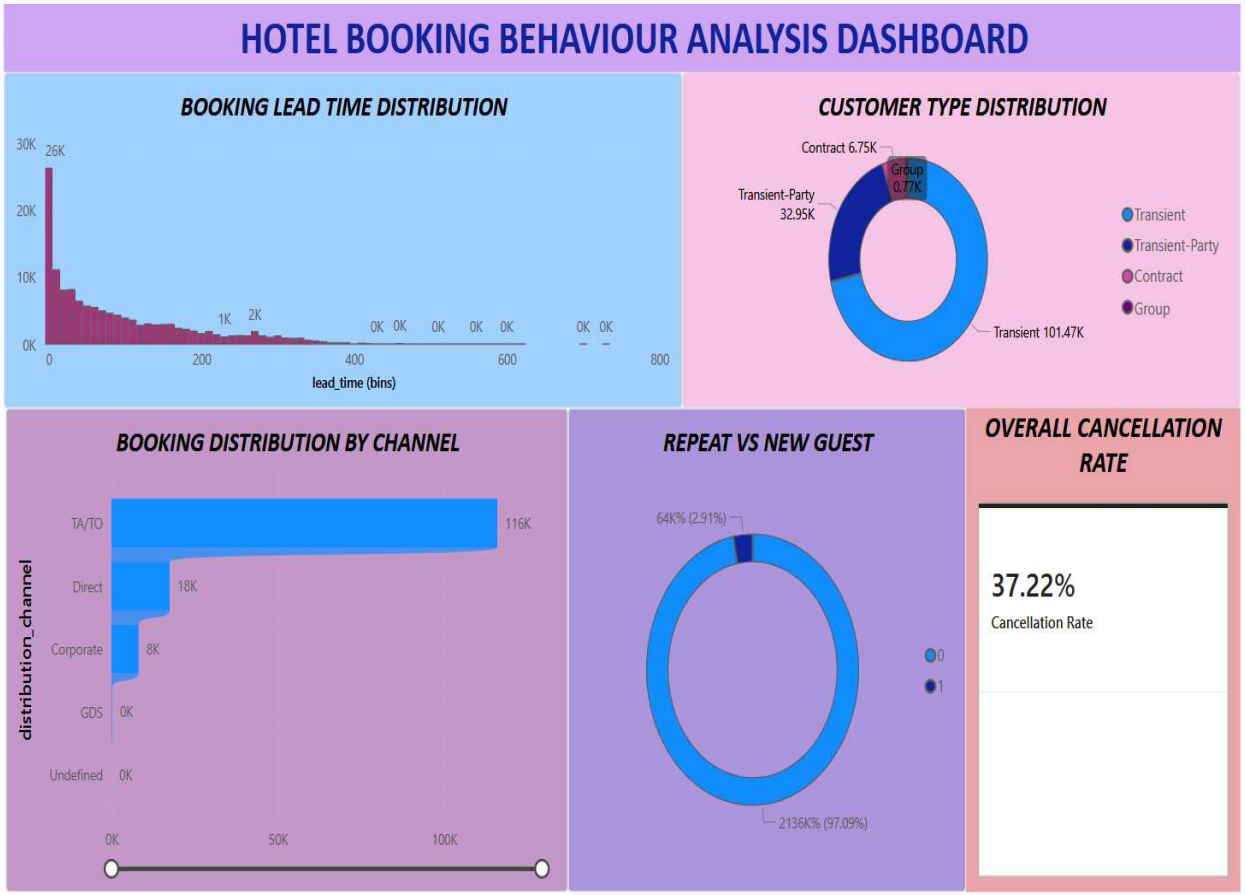
**86.8K**

Total Bookings

Volume of reservations processed

Key visualizations include monthly revenue trends and a breakdown of revenue by hotel type, offering a comprehensive view of the business.

# Dashboard 2: Booking Behavior Analysis




## Understanding Customer Booking Patterns

This dashboard focuses on how and when guests make reservations, providing critical insights for optimizing marketing and inventory management.

- Lead Time Distribution
- Booking by Customer Type (e.g., individual, corporate, group)
- Booking by Channel (e.g., OTA, direct, GDS)
- Repeat vs. New Guests

These metrics are vital for refining customer and product strategies.

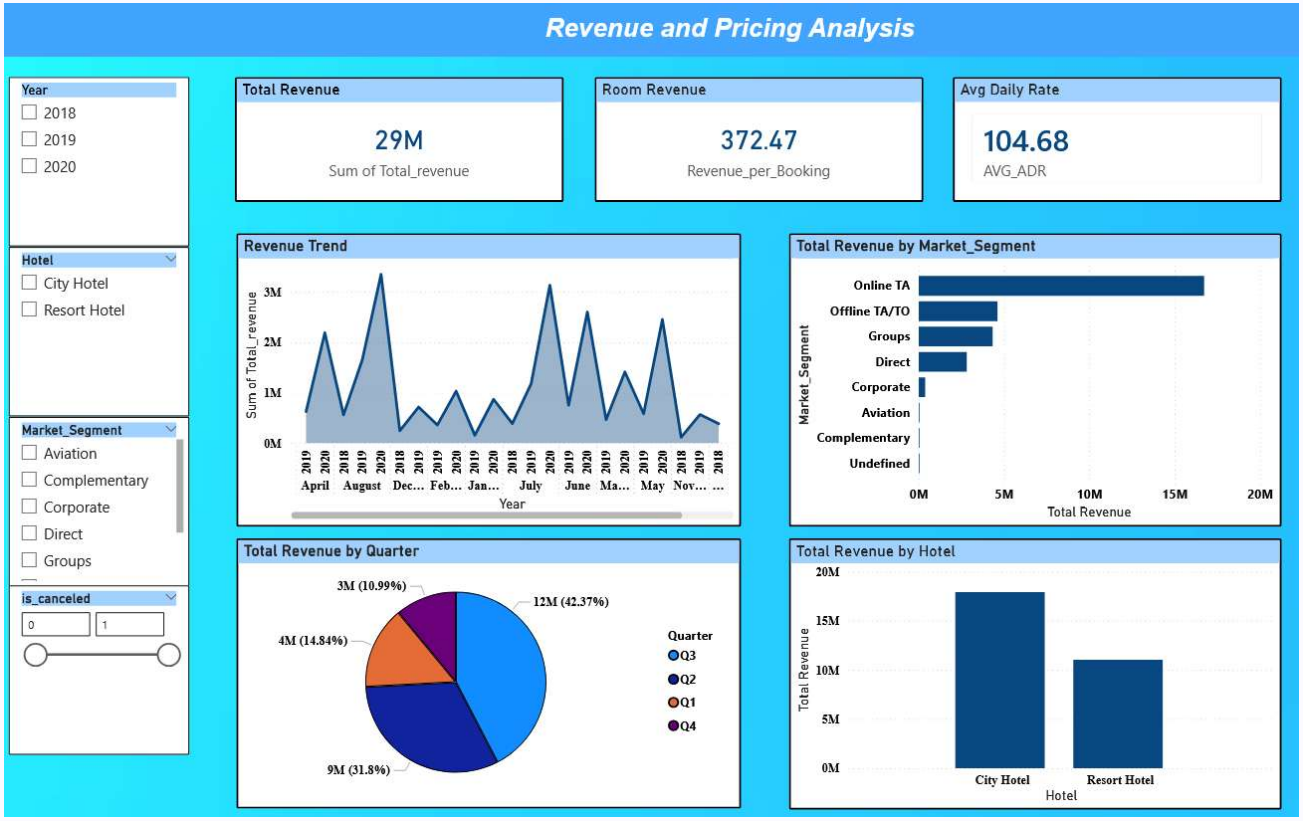
# Dashboard 3: Revenue & Pricing Analysis

## Analyzing Pricing Effectiveness and Revenue Drivers

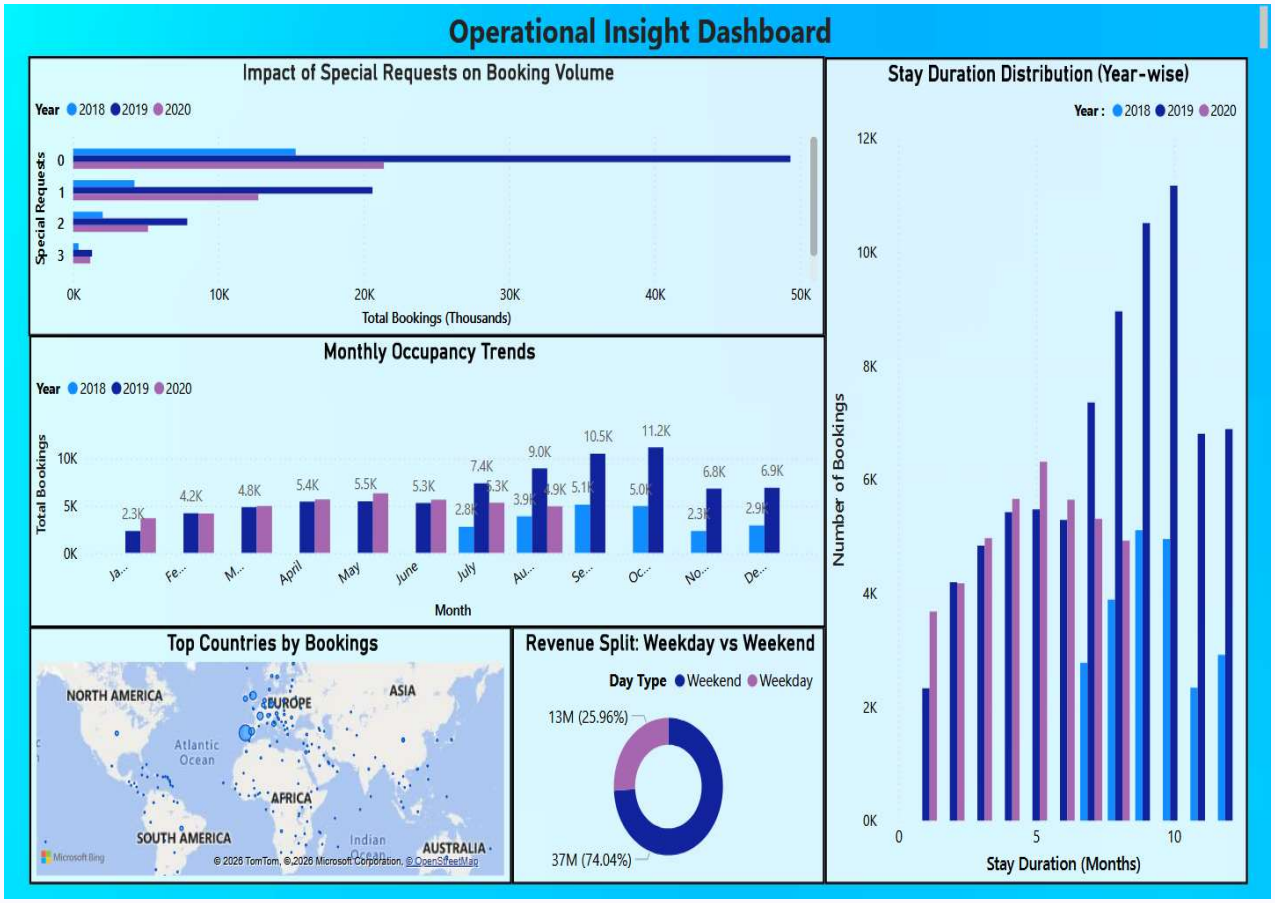
This dashboard dissects pricing strategies and their impact on revenue, helping identify profitable segments and seasonal trends.

- ADR (Average Daily Rate) Trend
- Revenue by Room Type
- Revenue by Season
- Revenue vs. ADR (Scatter Plot for correlation)
- Revenue per Booking

Key insights include identifying which room types generate the most revenue, the direct impact of pricing decisions, and seasonal performance variations.



# Dashboard 4: Operational Insights




## Understanding Operational Performance

This dashboard provides a comprehensive view of operational efficiency, guest demographics, and how these factors influence revenue.

- Stay Duration Distribution
- Weekday vs. Weekend Revenue
- Special Requests Impact on satisfaction and revenue
- Occupancy by Month
- Top Countries of Origin (Mapped)

It reveals operational trends, guest demographics, and differences in revenue patterns, showcasing strong business intelligence thinking.

# Business Impact: Driving Strategic Growth

### Improve Revenue Strategy

Leverage data-driven insights to develop and refine effective revenue generation plans.

### Reduce Cancellations

Identify patterns and implement proactive measures to minimize revenue loss from cancellations.

### Optimize Pricing

Dynamically adjust pricing based on demand, seasonality, and competitor analysis to maximize ADR and occupancy.

### Improve Occupancy Rate

Strategically manage inventory and marketing efforts to achieve higher occupancy across all periods.

### Support Management Decision-Making

Provide executives with clear, actionable intelligence for informed and timely strategic decisions.

# Conclusion: A 360° View for Success

The Hotel Revenue Analysis Dashboard system offers an unparalleled, complete 360° view of critical business dimensions:

- **Revenue Performance:** Deep dives into financial metrics and profitability.
- **Customer Behavior:** Understanding guest patterns from booking to stay.
- **Pricing Strategy:** Evaluating and refining competitive pricing models.
- **Operational Insights:** Optimizing daily processes and resource allocation.

This project demonstrates strong skills in Data Analytics, Business Intelligence, Dashboard Design, and Product Thinking, empowering hotels to thrive in a dynamic market.